

***** ==> REDUCE is over. Your final matrix is:

A =

1	0	0	$1/20$
0	1	0	$-3/5$
0	0	1	$3/2$

ans =

1.0000	0	0	0.0500
0	1.0000	0	-0.6000
0	0	1.0000	1.5000

>> rref()

Not enough input arguments.

Error in rref (line 23)

[m,n] = size(A);

>> a1 = [1 2 3 1; 1 3 0 1; 1 0 2 1]

a1 =

1	2	3	1
1	3	0	1
1	0	2	1

>> a2 = [8 7 3]'

a2 =

8
7
3

>> augA = [a1 a2]

augA =

1	2	3	1	8
1	3	0	1	7
1	0	2	1	3

>> rref(augA)

ans =

1	0	0	1	1
0	1	0	0	2
0	0	1	0	1

>> b1 = [1 1 3 -3; 0 2 1 -3; 1 0 2 -1]

```
b1 =
```

```
    1    1    3   -3
    0    2    1   -3
    1    0    2   -1
```

```
>> b2 = [0 3 -1]'
```

```
b2 =
```

```
    0
    3
   -1
```

```
>> augB = [b1 b2]
```

```
augB =
```

```
    1    1    3   -3    0
    0    2    1   -3    3
    1    0    2   -1   -1
```

```
>> rref([b1 b2])
```

```
ans =
```

```
    1    0    0    1    1
    0    1    0   -1    2
    0    0    1   -1   -1
```

```
>> rref(augB)
```

```
ans =
```

```
    1    0    0    1    1
    0    1    0   -1    2
    0    0    1   -1   -1
```

```
>> A = [1 2; -2 1]
```

```
A =
```

```
    1    2
   -2    1
```

```
>> rref(A)
```

```
ans =
```

```
    1    0
```

```
0      1
```

```
>> A = [1 2 3; 4 5 6; 7 8 9]
```

```
A =
```

```
1      2      3
4      5      6
7      8      9
```

```
>> rref(A)
```

```
ans =
```

```
1      0     -1
0      1      2
0      0      0
```

```
>> A = [1 2 3; 4 5 6; 7 8 0]
```

```
A =
```

```
1      2      3
4      5      6
7      8      0
```

```
>> rref(A)
```

```
ans =
```

```
1      0      0
0      1      0
0      0      1
```

```
>> A = [1 3; 1 2]
```

```
A =
```

```
1      3
1      2
```

```
>> rref([A eye(size(A))])
```

```
ans =
```

```
1      0     -2      3
0      1      1     -1
```

```
>> A = [1 1 2; 2 1 1; 1 2 1]
```

```
A =
```

```
1     1     2
2     1     1
1     2     1
```

```
>> rref([A eye(size(A))])
```

```
ans =
```

```
Columns 1 through 5
```

```
1.0000     0     0 -0.2500     0.7500
     0 1.0000     0 -0.2500 -0.2500
     0     0 1.0000     0.7500 -0.2500
```

```
Column 6
```

```
-0.2500
 0.7500
-0.2500
```

```
>> A = [1 2; 2 -1]
```

```
A =
```

```
1     2
2    -1
```

```
>> charpoly(A)
```

```
ans =
```

```
1     0    -5
```

```
>> A = [2 4 0; 1 2 1; 0 4 1]
```

```
A =
```

```
2     4     0
1     2     1
0     4     1
```

```
>> A = [2 4 0; 1 2 1; 0 4 2]
```

```
A =
```

```
2     4     0
1     2     1
0     4     2
```

```
>> charpoly(A)
```

ans =

1 -6 4 8

>> A = [1 0 0 0; 2 -2 0 0; 0 0 2 -1; 0 0 -1 2]

A =

1 0 0 0
2 -2 0 0
0 0 2 -1
0 0 -1 2

>> charpoly(A)

ans =

1 -3 -3 11 -6

>> A = [1 -3; 3 -5]

A =

1 -3
3 -5

>> e = eig(A)

e =

-2.0000
-2.0000

>> p = poly(e)

p =

1.0000 4.0000 4.0000

>> r = roots(p)

r =

-2.0000 + 0.0000i
-2.0000 - 0.0000i

>> A = [3 -1 4; -1 0 1; 4 1 2]

A =

```
    3    -1    4
   -1     0    1
    4     1    2
```

```
>> e = eig(A)
```

```
e =
```

```
 -2.3715
  0.8392
  6.5324
```

```
>> p = poly(e)
```

```
p =
```

```
 1.0000  -5.0000 -12.0000  13.0000
```

```
>> r = roots(p)
```

```
r =
```

```
 6.5324
 -2.3715
  0.8392
```

```
>> A = [2 -2 0; 1 -1 0; 1 -1 0]
```

```
A =
```

```
    2    -2     0
    1    -1     0
    1    -1     0
```

```
>> e = eig(A)
```

```
e =
```

```
    0
    0
    1
```

```
>> p = poly(e)
```

```
p =
```

```
    1    -1     0     0
```

```
>> r = roots(p)
```

```
r =
```

```
0
0
1
```

```
>> A = [2 4; 3 6]
```

```
A =
```

```
2    4
3    6
```

```
>> e = eig(A)
```

```
e =
```

```
0
8
```

```
>> p = poly(e)
```

```
p =
```

```
1    -8     0
```

```
>> r = roots(p)
```

```
r =
```

```
0
8
```

```
>> load cities.mat
```

```
>>
```